

Samad Mehndi

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EDUCATION

University of Texas at Dallas

MS Computer Engineering w/ Machine Learning || GPA – 3.82

Key Courses: Data Structures, Algorithms, Machine Learning, NLP, Computer Architecture, Db Design

Richardson, TX

May 2026

Guru Gobind Singh Indraprastha University

B.Tech in Electrical and Electronics Engineering

Delhi, India

Sep 2020

SKILLS

Languages: Python, Scala, C/C++, JavaScript, Java, SQL, NoSQL, PHP

Backend & APIs: Django, RESTful APIs, FastAPI, Flask, Microservices

Frontend: Vue.js, HTML5, CSS3, jQuery, AJAX

Databases: PostgreSQL, MySQL, MongoDB, Oracle, SQL Server

OS & Scripting: Linux (Ubuntu/CentOS/RedHat), Windows Server, PowerShell, Bash, Command Prompt, Batch Scripting

Tools & Core Strengths: Git, GitHub, Kubernetes (K8s), Docker, Jenkins, CI/CD, Postman, Jira, OOP, DSA, debugging/troubleshooting, communication, teamwork, ETL, Hadoop

WORK EXPERIENCE

Think Future Technologies

Gurugram, India

Software Engineer

Aug 2021 – Apr 2024

- Designed and maintained RESTful APIs for data ingestion, validation, and transformation workflows, handling 20K+ requests/day with secure authentication (JWT/OAuth2) and robust input validation
- Refactored backend services into modular, reusable components, reducing code duplication by ~30% and improving maintainability of scalable data-centric systems
- Delivered end-to-end features across large-scale web applications, collaborating with product, QA and DevOps to ship ~4 production releases per month
- Optimized backend performance by improving MongoDB indexing, aggregation pipelines, and query patterns, reducing API response time by ~40%
- Owned debugging and production support, resolving ~20 issues/month through log analysis, root-cause identification, and test-backed fixes
- Contributed to CI/CD pipelines using Git and Jenkins, enabling automated builds, testing, and reliable deployments across environments

Software Engineering Intern

May 2021 - July 2021

- Developed a movie listing web application with core features such as browsing and filtering, applying structured testing to validate functionality
- Contributed to backend development by implementing REST APIs for email notifications, camera alerts, and authentication/authorization, and built real-time notification delivery using WebSockets while debugging issues via log analysis and test cases

Coding Ninjas

Gurugram, India

Teaching Assistant

Aug 2020 – Dec 2020

- Solved 500+ debugging and algorithmic issues, strengthening structured problem-solving skills
- Explained low-level execution behavior and performance tradeoffs clearly to students

PROJECTS

JobMap AI – A Geospatial Job Intelligence Platform

Jun 2026

- Built and deployed a full-stack platform that maps live city-wise job postings geographically, replacing paginated list-scrolling with location-based discovery for job seekers, now live in production with Next.js, FastAPI, and PostgreSQL (Vercel, Railway, Neon)
- Designed LangGraph agent pipelines that extract structured fields from messy posting text and score listings against a user's uploaded resume, with pgvector semantic search (HNSW indexing) surfacing relevant roles in under a second despite wording mismatches

Wellness Search Engine

May 2026

- Built an end-to-end semantic search engine over a 143K-page wellness corpus, spanning indexing, retrieval, ranking, and a three-panel web UI that benchmarks results side by side against live Google and Bing
- Combined hybrid BGE-M3 dense and sparse retrieval over FAISS HNSW indexes with cross-encoder reranking and PageRank/HITS authority signals, KMeans topic clustering, and LLaMA-driven query expansion and answer generation, returning ranked results in ~0.8s across the full collection

AI-Assisted GPU Kernel Generation

Sep 2025

- Implemented Triton GPU kernels (deterministic and atomic-based) to analyze floating-point nondeterminism in parallel reductions
- Measured numerical drift using max ULP difference, observing up to ~34 ULP variation across configurations with varying concurrency
- Developed a logging and benchmarking framework to correlate execution configuration, performance, and numerical stability

Kidney Disease Classification

May 2025

- Built an end-to-end ML pipeline using TensorFlow and Keras to classify kidney disease from CT scan images, achieving 87.1% validation accuracy with 0.30 loss
- Integrated DVC for data versioning and MLflow for experiment tracking, with automated evaluation and deployment via Flask on AWS EC2
- Designed modular training and evaluation stages with YAML-based configuration and tracked pipeline reproducibility using DVC and Git